

JKBOSE 2026

CLASS 11TH

Bhaiicharya
on
TOP.

STRUCTURE OF ATOM

LECTURE- 09

CHEMISTRY

VISHAL SIR

JKBose Wallah



Topics To Covered

- 1 RULES OF FILLING ORBITALS**
- 2 ELECTRONIC CONFIGURATION OF ELEMENTS**
- 3 ~~X~~ ELECTRONIC CONFIGURATION OF ELEMENTS**

REVISION.

ORBITAL



- $\bar{e} \rightarrow$ probability (90%).

SHAPES OF ORBITAL.

a) s-orbital $\rightarrow$ spherical.

b) p-orbital $\rightarrow$ dumbbell shape
(p_x, p_y, p_z)

c) d-orbital $\rightarrow$ double-dumbbell
 $d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$

Non-Axial

Axial

doughnut

Radial + Angular node

$$\text{Radial node} = n - l - 1$$

$$\text{Angular node} = l$$

$$\text{Total no. of nodes} = \underline{n - 1}$$

Calculate the Radial, angular & total nodes for $4f$? $n=4$
 $\rightarrow l=3$

• Radial $\rightarrow n-l-1 = 4-3-1 = \underline{\text{ZERO}}$.

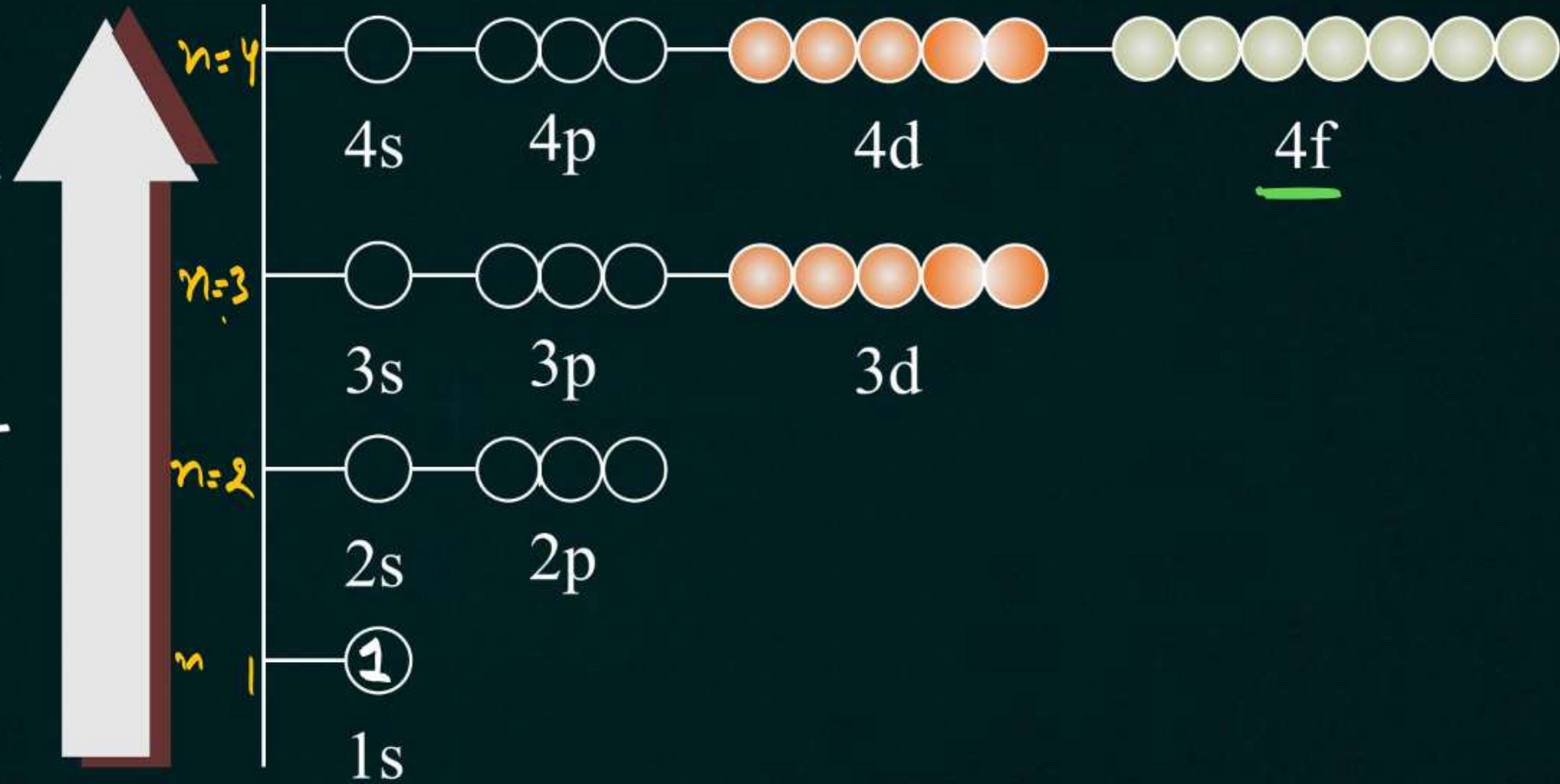
• Angular $\rightarrow l = 3$

• Total nodes $\rightarrow n-1 = 4-1 = 3$.

$l=0$	1	2	3
s	p	d	f

ENERGY LEVEL Diagram for hydrogen atom.

- For hydrogen energy of the orbital depends upon the value of 'n'
- Orbitals having same no. of 'n' have same energy
- Orbitals have same energy are degenerate orbitals.



$$5s = 5p = 5d = 5f = 5g$$

Energies of different orbitals in a hydrogen atom

Q. For hydrogen atom, the no. of degenerated orbitals in $n=4$.

a) 4.

b) 16

c) 9

d) 1

Total orbitals in any shell = n^2 .

#RULES OF FILLING ORBITALS.

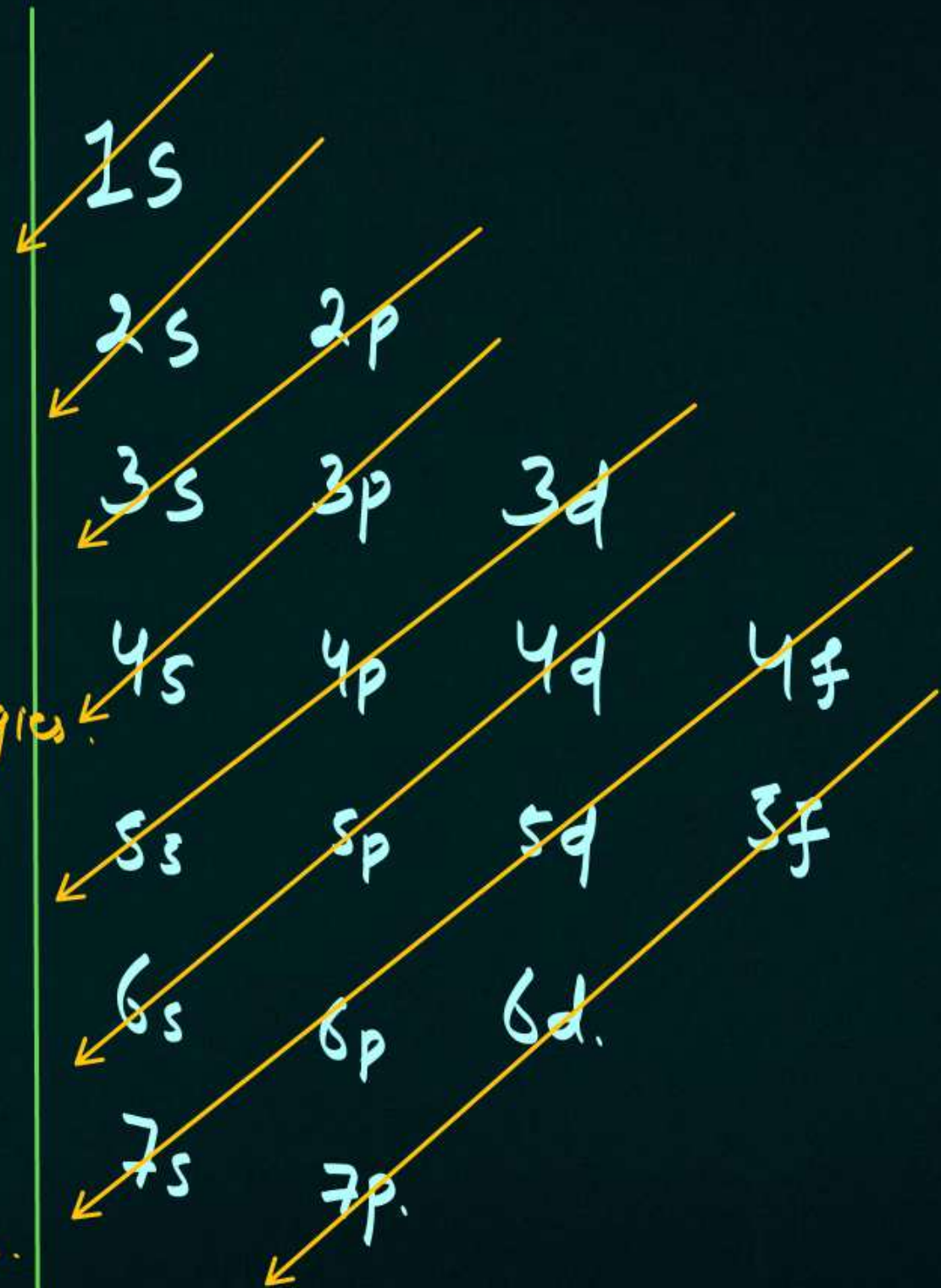
A) AUFBAU PRINCIPLE →

Aufbau is a German word, meaning "building up".

It states that "In the ground state of an atom electrons are filled according to increasing energies of orbitals. In other words, electrons first occupy the lowest energy orbital & then occupy the higher energy orbital available".

Order of energy of orbital

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p \dots$$



Energy of orbitals is determined by Bohr-Bury rule
OR $(n+l)$ rule $\rightarrow$

i) More the value of $(n+l)$, more is the energy.

e.g. $4s$, $(n+l) = 4+0 = \underline{4}$

$3d$, $(n+l) = 3+2 = \underline{5}$, then $3d$ has more energy than $4s$.

ii) If $(n+l)$ value, then more the value of 'n' more is the energy.

e.g. $2p$, $(n+l) = 2+1 = 3$

$3s$, $(n+l) = 3+0 = 3$, $3s$ has larger value of 'n' hence $3s$ has more energy than $2p$.

B) Pauli exclusion Principle $\rightarrow$

It states that "an orbital can occupy maximum of two electron & those of opposite spin".

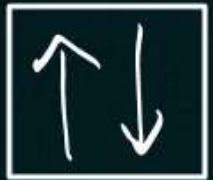
OR



No two electrons in an orbital can have same value of all the four quantum numbers.

i.e. These two electrons can have same value of three quantum no's (n, l, m_l) but can't have same value of spin quantum numbers.

1s



$n = 1$	1
$l = 0$	0
$m_l = 0$	0

$$m_s = +\frac{1}{2}$$

$$-\frac{1}{2}$$

C) HUND'S RULE OF MAXIMUM MULTIPLICITY $\rightarrow$

It states that "pairing of electrons in orbitals of same subshell (s, p, d etc) does not take place until each orbital of that subshell is singly occupied".

For example p^3 can be written in following way.

$\boxed{\uparrow \uparrow \uparrow}$, this correct representation

$$\text{Multiplicity} = +\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$\boxed{\uparrow \downarrow \uparrow}$, wrong.

$$\text{Multiplicity} = +\frac{1}{2} - \frac{1}{2} + \frac{1}{2} = \frac{1}{2}$$

$\boxed{\uparrow \downarrow \uparrow}$

$$\text{Multiplicity} = +\frac{1}{2} - \frac{1}{2} + \frac{1}{2} = \frac{1}{2}$$

Aufbau = $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p$

Paulis = $\boxed{\uparrow \downarrow}$

Munds = $\boxed{\uparrow \uparrow \uparrow} \quad p^3$

$\boxed{\uparrow \downarrow \uparrow} \quad p^3$

YAAD RAKHNA ✓.

S-Subshell $\rightarrow$ One orbital. $= 2\bar{e}$

p-subshell $\rightarrow$ three orbital (p_x, p_y, p_z) $= 6\bar{e}$

d-subshell $\rightarrow$ 5-orbital $= 10\bar{e}$

f-subshell $\rightarrow$ 7-orbital $= 14\bar{e}$
 $d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$

QUESTION



How many electrons can be present in a subshell with $l = 2$?

1 10

2 6

3 8

4 2

QUESTION

Calculate the $(n + l)$ value for a 5f orbital.

1 5

2 6

3 7

4 8

$$5f, (n+l) = 5+3 = 8$$



$$l = 0, 1, 2, 3$$

s p d f

QUESTION



For the principal quantum number $n = 3$, what is the maximum number of electrons that can be accommodated?

1 8

2 18

3 32

4 10



$$\begin{aligned}\underline{3 \text{ shell}} &= 2n^2 \\ &= 2 \times 3^2 \\ &= 18\end{aligned}$$

QUESTION

Which principle states that in the ground state of an atom, orbitals are filled in the order of their increasing energies?

- 1 Pauli Exclusion Principle
- 2 Hund's Rule of Maximum Multiplicity
- 3 Aufbau Principle
- 4 Heisenberg Uncertainty Principle

QUESTION

According to the Pauli Exclusion Principle, what is the maximum number of electrons an orbital can hold, and what must be their spin?

- 1 One electron, same spin
- 2 Two electrons, opposite spin
- 3 Two electrons, same spin
- 4 Four electrons, opposite spin

QUESTION

Which of the following is the correct order of filling orbitals?

1 $1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p$

2 $1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p$

3 $1s, 2s, 2p, 3s, 3d, 3p, 4s, 4p$

4 $1s, 2s, 3s, 2p, 3p, 3d, 4s$

- ① Aafreena Fayaz
- ② Shah Abdullah.
- ③ Musaa.

IUV 

Bhauichara On Top.

REVISE KO Aadat Banalo.

Thank
You